

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Physics) Examination March 2018

PD203 Digital Systems and Applications

Date: 28.03.2018, Day: Wednesday Time: 01:30 p.m. To 02:00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q-I Choose the correct answer for the following questions. (Each question carries one mark). [20]

1. The value of base r if $(121)_r = (144)_8$
(a) 7 (b) 8
(c) 9 (d) 10
2. _____ is an active component.
(a) Resistor (b) Diode
(c) Inductor (d) Capacitor
3. Number of transistors in LSI Chip is _____
(a) 1 to 10 (b) 10 to 500
(c) 500 to 20,000 (d) 20,000 to 10,00,000
4. When used with an IC, what does the term "QUAD" indicates?
(a) 2 Circuits (b) 6 Circuits
(c) 4 Circuits (d) 8 Circuits

5. $A + BC =$ _____
- (a) BC (b) $(A + B)(A + C)$
(c) $AB + AC$ (d) $(A + B)C$
6. $\sum m(0,1,2,3) =$
- (a) $\bar{A}$ (b) B
(c) 1 (d) 0
7. The output of a logic gate is '1' when all its inputs are at logic 0. The gate is either
- (a) NAND or an EX-OR gate (b) an OR or an EX-NOR gate
(c) NOR or an EX-NOR gate (d) an AND or an EX-OR gate
8. A group of 4 bit is called ____.
- (a) bits (b) byte
(c) nibble (d) ripple
9. How many flip flops requires to make a mod-32 binary counter?
- (a) 3 (b) 45
(c) 5 (d) 6
10. A master slave flip flop has the characteristic that
- (a) Change in the input immediately reflected in the outputs
(b) Change in the output occurs when the state of master is affected
(c) Change in the output occurs when the state of the slave is affected
(d) Both the master and slave states are affected at the same time
11. A circuit that converts n inputs to 2^n outputs is called _____
- (a) Encoder (b) Decoder
(c) Comparator (d) Counter
12. Process of converting data into a form that can be easily understood by people is called ____.
- (a) Translating (b) Data organization
(c) Encoding (d) Decoding
13. 555 timer is ____ pin IC?
- (a) 8 (b) 10
(c) 14 (d) 16

14. How many bits is 8085 microprocessor?
- (a) 8 (b) 16
(c) 256 (d) 512
15. Which company is the biggest player in the microprocessor industry?
- (a) Motorola (b) IBM
(c) Intel (d) AMD
16. Decoder is a _____.
- (a) combinational circuit (b) sequential circuit
(c) complex circuit (d) gate
17. Indicate which of the following logic gates can be used to realize all possible combinational Logic functions.
- (a) OR gates only (b) NAND gates only
(c) EX OR gates only (d) NOT gates only
18. Based on how binary information is entered or shifted out, shift registers are classified into ____ categories.
- (a) 2 (b) 3
(c) 4 (d) 5
19. Latches constructed with NOR and NAND gates tend to remain in the latched condition due to which configuration feature?
- (a) Low input voltage (b) Synchronous operation
(c) Gate impedance (d) Cross coupling
20. A bulb in staircases has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by and one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles _____.
- (a) an AND gate (b) an OR gate
(c) an XOR gate (d) a NAND gate

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Date: 28.03.2018, Day: Wednesday Time: 02:00 p.m. To 04:30 p.m. Maximum Marks: 50

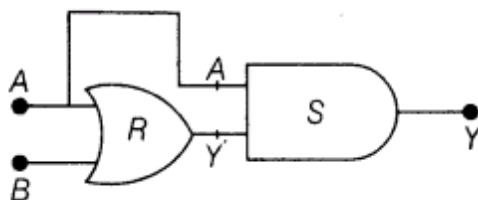
Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

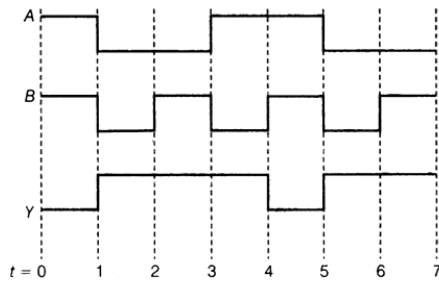
SECTION – I

Q – II Answer any TEN of the following questions. (Each question carries two marks.) [20]

- a) What is the difference between positive edge triggered, negative edge triggered and level triggered circuits.
- b) Give examples of 1- byte instruction and explain them.
- c) Draw the block diagram of CRO.
- d) What are the various registers in microprocessor 8085?
- e) Subtract using 9's and 10's complement: $416.73 - 2928.54$
- f) Prove $AB + \bar{A}C + BC = AB + \bar{A}C$
- g) Explain: Why is the binary number system used in digital system? Convert $(52.75)_{10}$ into binary.
- h) Draw the block diagram of RS and D flipflop.
- i) Write the truth table for the combination of the gates shown. Name the gates used.



- j) The input wave forms A and B and the output waveform Y of a gate are shown below. Name the gate it represents, write its truth table and draw the logic symbol of this gate.



- k) Name the 5 flags in 8085 microprocessor.
- l) Convert $(18)_{10}$ into binary and octal number system.
- m) Convert $(22)_8$ into decimal number system.

SECTION – II

Q-III Answer any SIX of the following questions. (Each question carries five marks.) [30]

- a) Design: 2 bit synchronous up counter.
- b) Draw and explain the circuit diagram of IC 555 in monostable mode.
- c) Draw the circuit diagram and explain the truth table of Half adder and half subtractor.
- d) Reduce the expression $f = \sum m(0,2,3,4,5,6)$ using mapping and implement it using logic gates.
- e) Realize the basic gates using the universal gates. Draw the appropriate circuits diagrams and verify the truth tables.
- f) State and prove both the De Morgan's law.
- g) Draw the circuit diagram of JK flip flop and explain it with the truth table.
- h) Design: BCD asynchronous up counter.